

EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)
Homework 2 – Sensitivity of Nitrate Acid Production
Total Score: 50

Learning Objectives:

- Interpret a pressure-dependent rate expression with the knowledge of chemical kinetics.
- Modify code on a chemical mechanism correctly by identifying the specific reaction and parameters to change.
- Configure and run a sensitivity experiment and investigate the effect of changing the rate constant of a chemical production pathway.

Tropospheric HNO_3 is produced mainly by two pathways, gas-phase $\text{OH} + \text{NO}_2$ during daytime and heterogeneous N_2O_5 hydrolysis during nighttime. We consider how sensitive is total HNO_3 production to the rate constant of the dominant daytime pathway. This homework assignment invites you to use those mechanics to run a controlled sensitivity experiment on the $\text{OH} + \text{NO}_2$ rate constant, and to interpret the result.

We simulate GEOS-Chem over the period 2019-07-01 00:00 UTC to 2019-07-08 00:00 UTC over the globe. We then use the bounding box (LON = [115, 118], LAT = [38, 41]) to focus entirely on the Beijing region. Daytime is defined as the period over local-time hours 06:00-18:59.

Part 1. Baseline Analysis

Question 1 (8 pts). Complete the table below for the Beijing box:

Pathway	Day total (molec/cm ³)	Night total (molec/cm ³)	Day fraction	Night fraction
$\text{NO}_2 + \text{OH}$				
N_2O_5 hydrolysis				
Other				
Sum			100%	100%

In a few sentences, explain why the fractions come out the way they do.

Question 2 (8 pts). Plot the three pathways as a time series over the 48-hour window. Shade daytime hours (06:00–18:59 local) in yellow. Label the axes with units. Save as `baseline_pathways.png` and include it in your report. **In 2 sentences, describe what the plot shows about the diurnal cycles of the three pathways.**

Part 2. Chemical Mechanism

Question 3 (10 pts). The rate expression for this reaction is:

$\text{NO}_2 + \text{OH} \{+M\} = \text{HNO}_3 \{+M\}$: GCJPLPR_aba(1.80d-30, 3.0d+00, 2.8d-11, 0.6d0);

Explain what each argument of GCJPLPR_aba means, and why this reaction needs two pre-factors rather than one.

Your answer should:

- Name all four arguments and state their physical units
- Explain what it means for a reaction to be "pressure-dependent"
- Explain which limit (low-pressure or high-pressure) dominates at the surface

Part 3. Sensitivity Runs

Question 4 (24 pts). Consider two cases concerning the rate constant of the reaction $\text{NO}_2 + \text{OH} \{+M\} = \text{HNO}_3 \{+M\}$, where the "low case" sets the rate constant of the reaction to exactly 50% of its baseline value at every pressure, and the "high case" sets the rate constant of the reaction to exactly 200% of its baseline value at every pressure. By running the two additional cases, complete the two comparison tables below. The 0.5× column refers to the low case, the 1.0× column to the baseline, and the 2.0× column to the high case; day and night refer to total HNO_3 production via a given pathway computed over local-time hours 06:00–18:59 and 19:00–05:59, respectively.

Day totals:

Pathway	0.5× day	1.0× day	2.0× day	Δ (0.5 vs 1) %	Δ (2.0 vs 1) %
$\text{NO}_2 + \text{OH}$					
N_2O_5 hydrolysis					
Other					
Total					

Night totals:

Pathway	0.5× night	1.0× night	2.0× night	Δ (0.5 vs 1) %	Δ (2.0 vs 1) %
$\text{NO}_2 + \text{OH}$					
N_2O_5 hydrolysis					
Other					
Total					

Describe what the two tables show. Does the modified pathway scale exactly with the multiplier, and how does the other pathways change and in which direction?

Did N_2O_5 hydrolysis change between the three runs? If yes, in which direction and by what percentage? If not, state that the change is smaller than your analysis can resolve.

Submission Guideline: To complete the lab, please submit **TWO** files:

- **Hw2_<your_id>.pdf**, which contains the answers to all the 4 questions;
- **hw2_<your_id>.zip**, which contains the Python scripts to produce the included plots and scripts that perform data analysis.

If any AI tool was used in writing or code generation, please include a short note on AI use.

----- End of Homework 2 -----